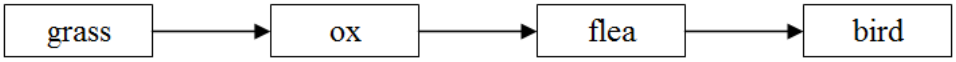
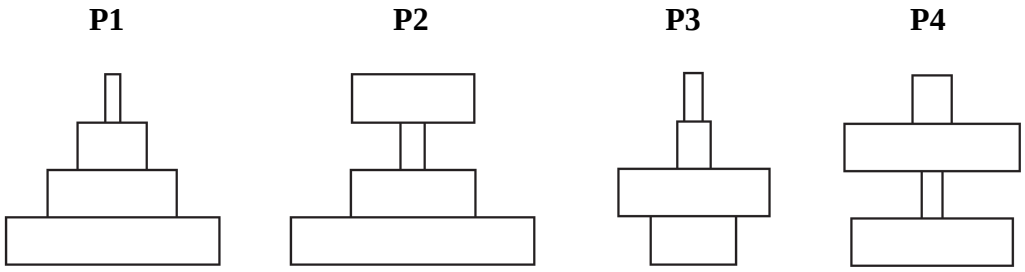
Multiple-choice questions

|!|BCEMB0200300001|!| ★

Below are a food chain on a grassland and four ecological pyramids (P1, P2, P3 and P4).





Which of the following combinations correctly matches the pyramid of numbers and the pyramid of biomass with the food chain?

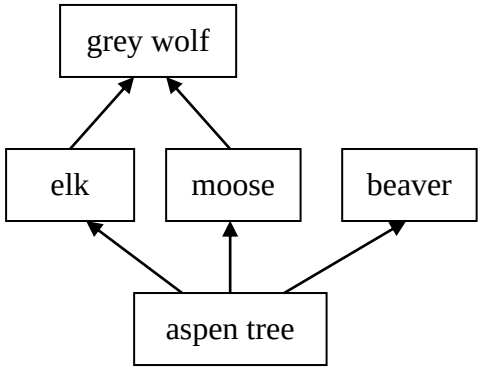
	Pyramid of numbers	*Pyramid of biomass*
A.	P2	P1
B.	P4	P1
C.	P3	P4
D.	P1	P4

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0200300002|!| ★

The following diagram shows part of a food web in a national park of North America.



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Which of the following would occur if the number of grey wolf decreased drastically?

(1) The number of elk would increase.
(2) The number of beaver would increase.
(3) The number of aspen tree would decrease.

A. (1) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

□

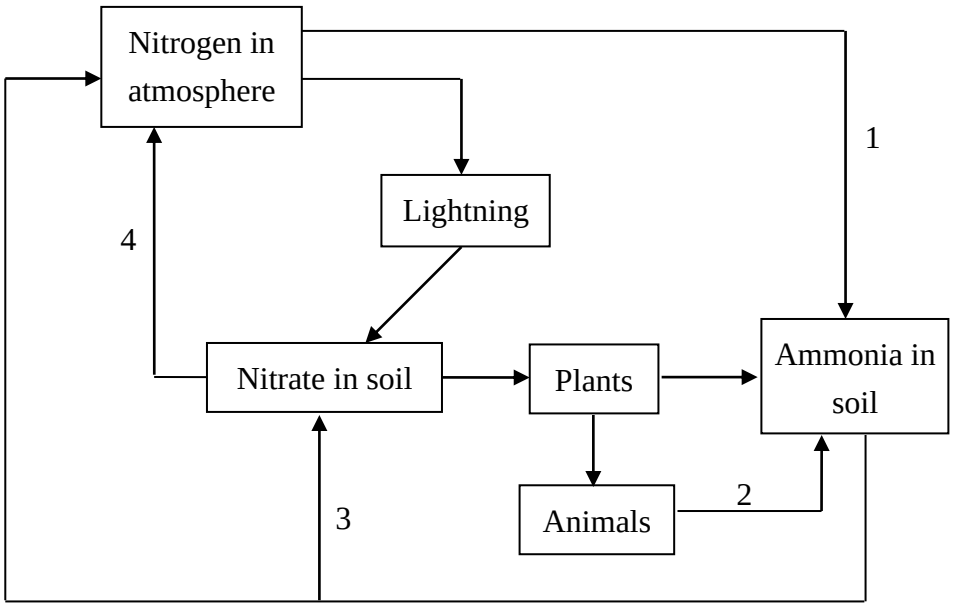
##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0200500003|!| ★

✂

Directions: The two following questions refer to the diagram below that shows a simplified nitrogen cycle.



Which are processes 1 and 4?

	Process 1	***Process 4***
A.	nitrification	nitrogen fixation
B.	nitrogen fixation	denitrification
C.	nitrification	denitrification
D.	nitrogen fixation	ammonification

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0200500004|!| ★

✂

Which of the following microorganisms are involved in processes 2 and 3, respectively?

	Process 2	***Process 3***
A.	putrefying bacteria	nitrifying bacteria
B.	denitrifying bacteria	nitrogen fixing bacteria
C.	nitrogen fixing bacteria	nitrifying bacteria
D.	nitrifying bacteria	nitrogen fixing bacteria

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0200100005|!|

Which of the following correctly arranges the different levels of organization in ecology in order of ascending scale?

A. species → community → population → biome
B. species → population → community → ecosystem
C. population → species → ecosystem → biosphere
D. population → community → biosphere → biome

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0200100006|!|

Which level of organization in ecology are tropical rainforests at?

A. population
B. community
C. biome
D. biosphere

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##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0200200007|!|

Which of the following is an / are adaptation(s) developed by certain animals that allow them to cope with cold climates?

(1) accumulation of thick layers of subcutaneous fat
(2) hibernation
(3) migration to warmer regions

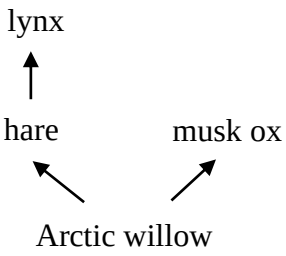
A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0200300008|!| ★

The following figure shows a food web in the Arctic region.



If the number of lynx is increasing, which of the following will occur as a result?

(1) The number of hare will increase.
(2) The number of Arctic willow will decrease.
(3) The number of musk ox will increase.

A. (2) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

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##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0200200009|!|

Which of the following pairs is an example of mutualism?

A. barnacles and crabs
B. algae and fungi in lichens
C. denitrifying bacteria and leguminous plants
D. tapeworms and humans

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0200300010|!| ★

The following diagram shows a pyramid of biomass.



Which of the following statements about the food chain represented by the pyramid of biomass is ***incorrect***?

A. The biomass of the producer is the greatest.
B. The biomass decreases as the trophic level increases.
C. The quaternary consumer is at the highest trophic level.
D. There are four trophic levels in the food chain.

##C##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0200300011|!| ★

Which of the following organisms can provide food for decomposers?

(1) producers
(2) consumers
(3) decomposers

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

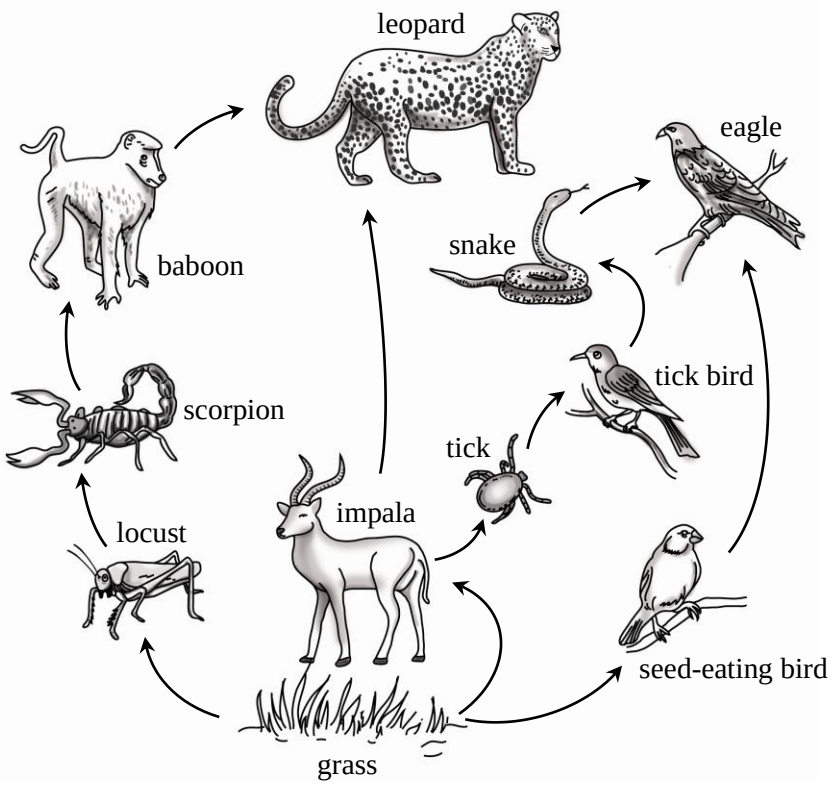
□

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0200300012|!| ★

Directions: The three following questions refer to the diagram below that shows a food web in an African grassland.



Which of the following organism(s) is a / are tertiary consumer(s) in the food web?

(1) eagle
(2) baboon
(3) tick bird

A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0200300013|!| ★

What is the maximum number of trophic levels found in the above food web?

A. 3
B. 4
C. 5
D. 6

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0200200014|!| ★

Based on the above food web, what is the ecological relationship between the impala and tick?

A. predation
B. competition
C. parasitism
D. mutualism

##C##

Skill assessed	Mark
Analysing data	1

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|!|BCEMC0200300015|!|

Which of the following statement(s) about the organisms in a food chain is correct?

(1) The producers are at the highest trophic level.
(2) Decomposers can feed on the dead remains of both producers and consumers.
(3) Secondary consumers feed on producers.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

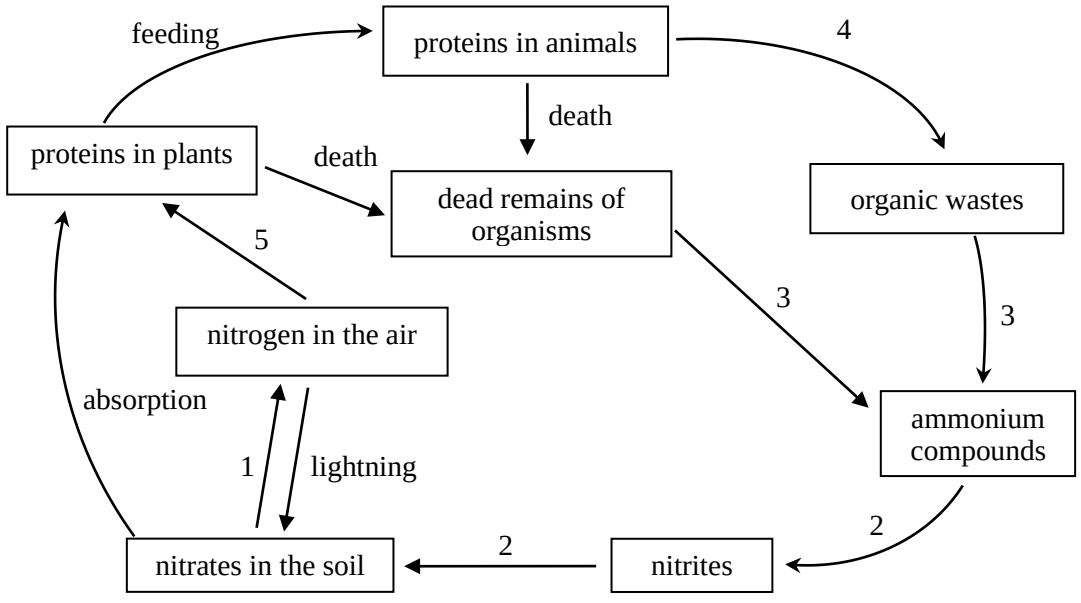
##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0200500016|!|

✂

Directions: The three following questions refer to the diagram below that shows a nitrogen cycle.



Process 3 represents

A. respiration.
B. photosynthesis.
C. nitrification.
D. decomposition.

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##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0200500017|!|

✂

Which of the following processes in the given diagram requires putrefying bacteria?

A. process 1
B. process 2
C. process 3
D. process 4

□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0200500018|!|

✂

Which of the following processes in the given diagram requires denitrifying bacteria?

A. process 1
B. process 3
C. process 4
D. process 5

□

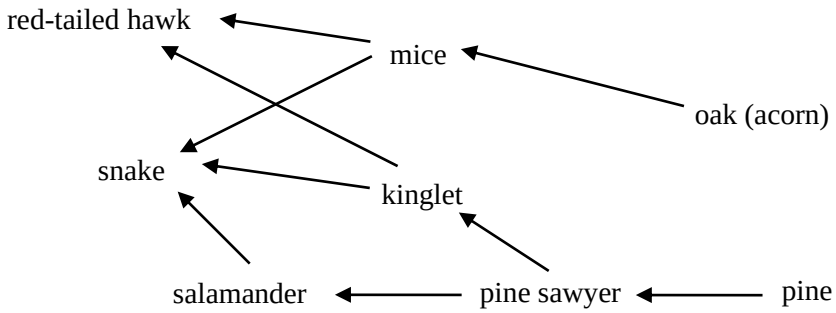
##A##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0200200019|!| ★

Directions: The two following questions refer to the figure below, which shows a food web in a forest.



What is the ecological relationship between the kinglet and salamander?

A. competition
B. mutualism
C. commensalism
D. predation

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0200300020|!| ★

The snake in the given food web is considered to be

(1) a primary consumer.
(2) a secondary consumer.
(3) a tertiary consumer.

A. (2) only
B. (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Analysing data	1

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|!|BCEMC0200300021|!|

Which of the following factor(s) will lead to energy loss in a food chain?

(1) uneaten organisms
(2) photosynthetic products
(3) excretory products

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0200200022|!|

Which of the following is / are abiotic factor(s) of a freshwater stream ecosystem?

(1) water salinity
(2) wind speed
(3) species diversity

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0200200023|!| ★

The photograph below shows an organism that lives in the Arctic region.



Which of the following is an / are the organism's adaptation(s) to the Arctic region?

(1) It has a large surface area to volume ratio to increase heat gain.
(2) It has dense fur to reduce heat loss.
(3) It has a thick layer of subcutaneous fat to reduce heat loss.

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0200200024|!|

Which of the following is the major factor that determines the number of plants present in a forest?

A. wind speed
B. temperature
C. carbon dioxide concentration in the air
D. light intensity

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0200300025|!|

Which of the following groups of organisms usually have the greatest biomass in an ecosystem?

A. primary consumers
B. producers
C. decomposers
D. secondary consumers

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0200300026|!|

Primary consumers of an ecosystem are usually

A. herbivores.
B. carnivores.
C. autotrophs.
D. omnivores.

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0200300027|!| ★

Which of the following is ***not*** an essential component of an ecosystem?

A. sunlight
B. decomposer
C. producer
D. secondary consumer

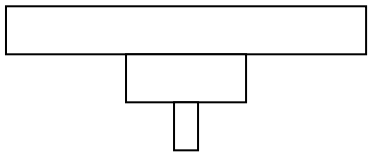
##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0200300028|!| ★

The following figure shows a pyramid of numbers of a habitat.



Which of the following food chains would be best represented by this pyramid of numbers?

A. microscopic algae → shrimp → fish
B. grass → cow → human
C. tree → caterpillar → bird
D. tree → bird → parasitic worm

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0200200029|!|

Which of the following activities of a flowering plant are affected by light?

(1) flowering
(2) formation of starch
(3) direction of shoot growth

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0200300030|!|

The producer of one food chain transfers less than 10% of its stored energy to primary consumers. Which of the following is the correct explanation for this?

A. Energy is used when primary consumers break down food.
B. Energy is used when the producer carry out photosynthesis.
C. Energy is lost as transpiration occurs in the producer.
D. Energy is lost as heat from the producer to the environment.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0200300031|!| ★

All food chains begin with

A. parasites.
B. saprophytes.
C. autotrophs.
D. heterotrophs.

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0200300032|!| ★

The fleas on the skin of a fox are ***most*** probably

A. producers.
B. primary consumers.
C. secondary consumers.
D. tertiary consumers.

##D##

Skill assessed	Mark
Applying concepts	1

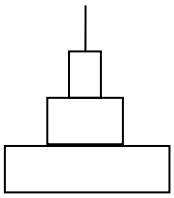
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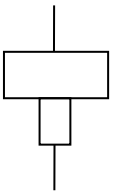
|!|BCEMB0200300033|!| ★

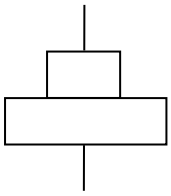
Directions: The two following questions refer to the food chain below.

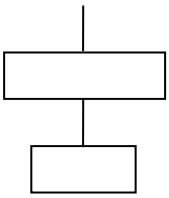
tree → aphid → bird → hawk

Which of the following pyramids of numbers best represents the food chain above?

A. 

B. 

C. 

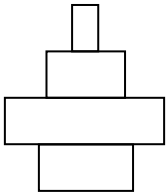
D. 

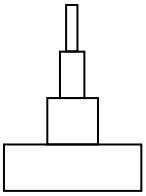
##C##

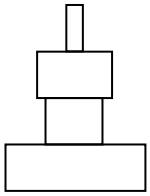
Skill assessed	Mark
Analysing data	1

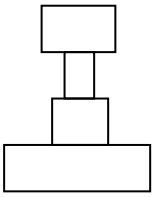
|!|BCEMB0200300034|!| ★

Which of the following pyramids of biomass best represents the food chain above?

A. 

B. 

C. 

D. 

##B##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0200300035|!| ★

The following shows a food chain.

pond weed → dragonfly nymph → frog → snake

Decomposers can feed on the dead remains of

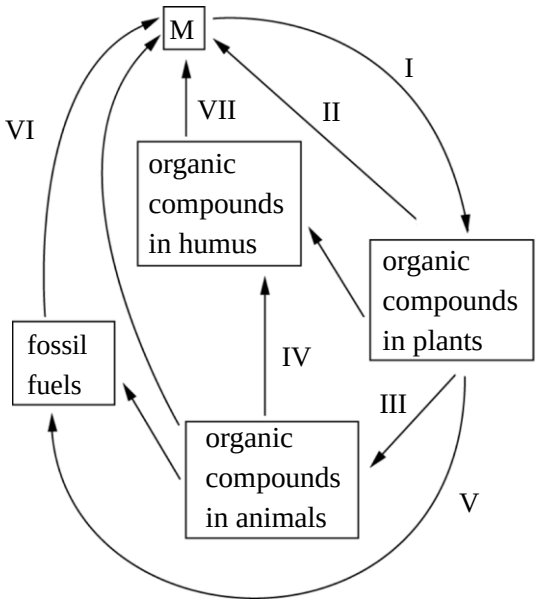
A. pond weeds only
B. snakes only
C. dragonfly nymphs, frogs and snakes only
D. pond weeds, dragonfly nymphs, frogs and snakes

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0200400036|!| ★

Directions: The three following questions refer to the diagram below that shows a carbon cycle.



M represents

A. the organic compounds in decomposers.
B. the carbon dioxide consumed during photosynthesis in plants.
C. the carbon dioxide produced during respiration in organisms.
D. the carbon dioxide in the atmosphere.

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##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0200400037|!| ★

Which of the following processes in the given diagram involve(s) decomposers?

(1) Process II
(2) Process IV
(3) Process V

A. (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0200400038|!| ★ **Cross topic**

Which of the following processes in the given diagram involves carbon fixation?

A. Process I
B. Process III
C. Process VI
D. Process VII

##A##

Skill assessed	Mark
Analysing data	1

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|!|BCEMC0200100039|!|

Which of the following can be found in an ecosystem?

(1) cycling of carbon
(2) cycling of energy
(3) interaction between organisms and the physical environment

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0200300040|!|

Which of the following trophic levels involve(s) consumers?

(1) first trophic level
(2) second trophic level
(3) top trophic level

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0200300041|!|

Which of the following statements about food chains is / are correct?

(1) All food chains must start with producers.
(2) Decomposers are never shown in food chains.
(3) The minimum number of organisms in a food chain is one.

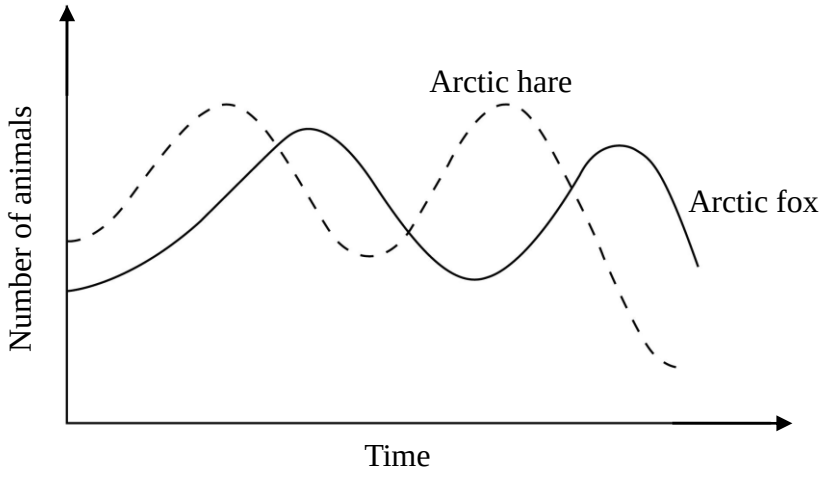
A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0200200042|!| ★

Directions: The three following questions refer to the graph below that shows the changes in the populations of the Arctic hare and Arctic fox over a period of time.



What is the ecological relationship between the Arctic hare and Arctic fox?

A. commensalism
B. competition
C. mutualism
D. predation

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0200200043|!| ★

Which of the following descriptions about the Arctic hare and Arctic fox is ***cannot*** be deduced from the graph above?

A. The Arctic hare has a higher reproductive rate than the Arctic fox.
B. The total biomass of the Arctic hare is greater than that of the Arctic foxes.
C. The Arctic fox is at a higher trophic level than the Arctic hare.
D. The Arctic hare has a longer lifespan than the Arctic fox.

☐

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0200100044|!| ★

In which biome do the Arctic hare and Arctic fox live?

A. desert
B. temperate grassland
C. tropical savanna
D. tundra

☐

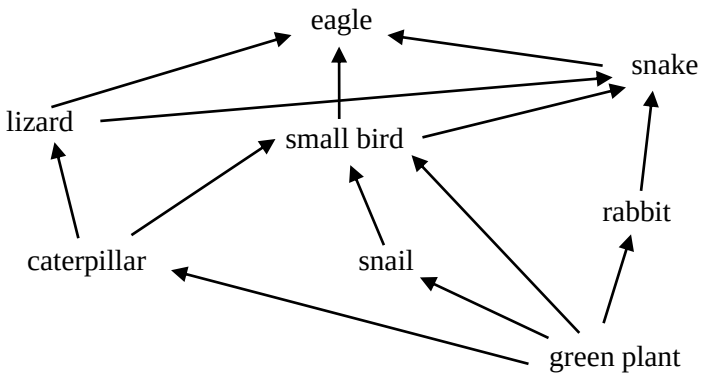
##D ##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0200300045|!| ★

Directions: The following two questions refer to the food web below.



Which of the following combinations correctly matches the organisms with their positions in the food chains?

	Primary consumer	***Secondary consumer***	***Tertiary consumer***
A.	caterpillar	small bird	lizard
B.	green plant	rabbit	snake
C.	eagle	snake	snail
D.	rabbit	lizard	eagle

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0200200046|!| ★

What is / are the ecological relationship(s) between the snake and eagle as shown in the food web?

(1) predation
(2) competition
(3) mutualism

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##B##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0200200047|!| ★

Which of the following is an example of parasitism?

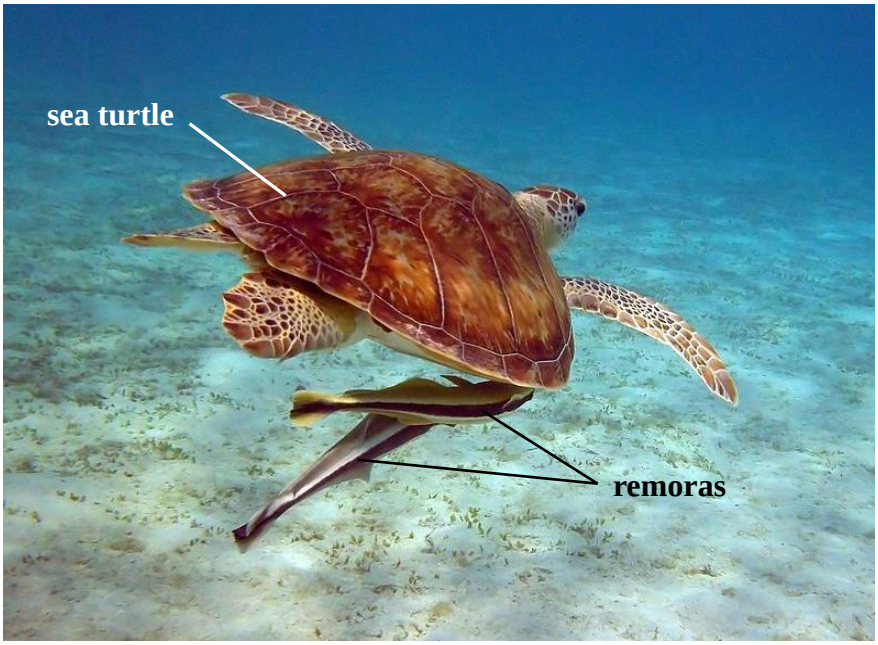
A. vitamin-producing bacteria living in the intestine of a human
B. barnacles and limpets living on a rock surface
C. sea anemone living on the shell of a hermit crab
D. tapeworms living in the intestine of a monkey

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0200200048|!| ★

The photograph below shows two remoras attached to a sea turtle.



(Image source: https://commons.wikimedia.org/wiki/File:Sea_Turtle_with_Remoras_-_Marsa_Alam,_Egypt_-_August_9,_2011.jpg)

In doing so, the remora can reduce the energy it expends to swim. It also receives food debris when the sea turtle feeds. The sea turtle is unaffected by its association with the remora. Thus the ecological relationship between the remora and the sea turtle is an example of

A. predation.
B. parasitism.
C. competition.
D. commensalism.

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##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0200200049|!|

Which of the following ecological relationships harm(s) both involved species?

(1) parasitism
(2) competition
(3) commensalism

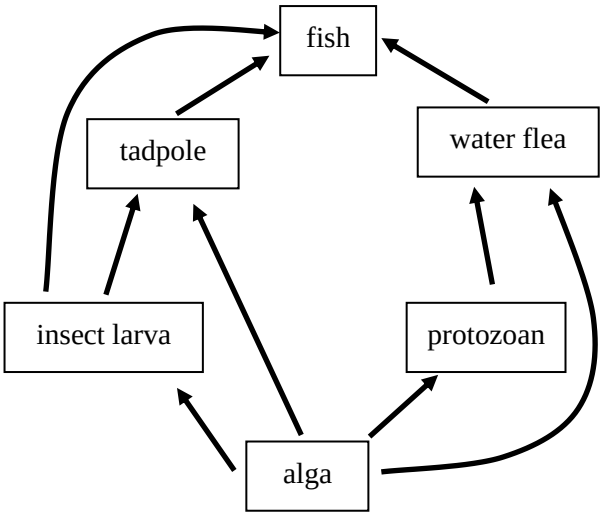
A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0200300050|!| ★

Directions: The following three questions refer to the food web below.



Which of the following animals are primary consumers?

A. tadpoles only
B. insect larvae and protozoa only
C. insect larvae, protozoa and tadpoles only
D. insect larvae, protozoa, tadpoles and water fleas only

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##D ##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0200200051|!| ★

Which of the following is the ecological relationship between tadpoles and water fleas as shown in the food web above?

A. predation
B. mutualism
C. intraspecific competition
D. interspecific competition

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0200300052|!| ★

Based on their diet, tadpoles can be classified as

A. herbivores.
B. carnivores.
C. omnivores.
D. autotrophs.

##C##

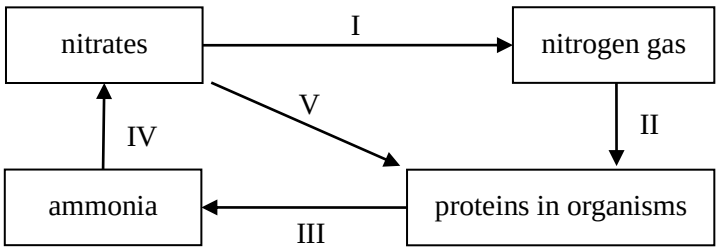
Skill assessed	Mark
Analysing data	1

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|!|BCEMB0200500053|!| ★

✂

Directions: The following three questions refer to the diagram below that shows a simplified nitrogen cycle.



Which process in the diagram occurs more rapidly in waterlogged soil than in well-drained soil?

A. I
B. II
C. III
D. IV

□

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0200500054|!| ★

✂

Which process in the diagram is nitrogen fixation?

A. I
B. II
C. III
D. V

□

##B##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0200500055|!| ★

✂

Which of the processes in the diagram involve bacteria?

A. I and II only
B. I, II and III only
C. I, II, III and IV only
D. I, II, III, IV and V

□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0200500056|!|

✂

Which of the following statements about nitrogen-fixing bacteria is / are correct?

(1) They are saprophytic organisms.
(2) They are present in the root nodules of leguminous plants.
(3) They can convert the nitrogen in soil gas into ammonium compounds.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

□

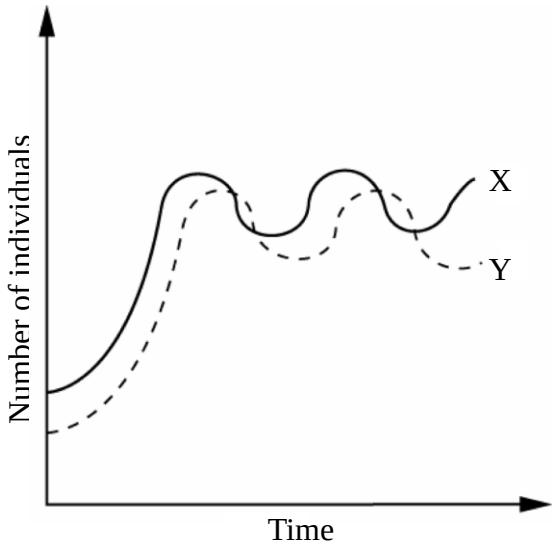
##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0200200057|!| ★

Directions: The two following questions refer to the graph below that shows the changes in the population sizes of two species (X and Y) in a habitat over a period of time.



Which of the following statements about the graph is / are correct?

(1) The peak of the population of X is higher than that of the population of Y.
(2) The population of Y changes with that of X but lags slightly behind.
(3) The population of X is directly proportional to the population of Y.

A. (1) only
B. (2) only
C. (1) and (2) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0200200058|!| ★

What is the ecological relationship between X and Y?

A. predation
B. competition
C. mutualism
D. parasitism

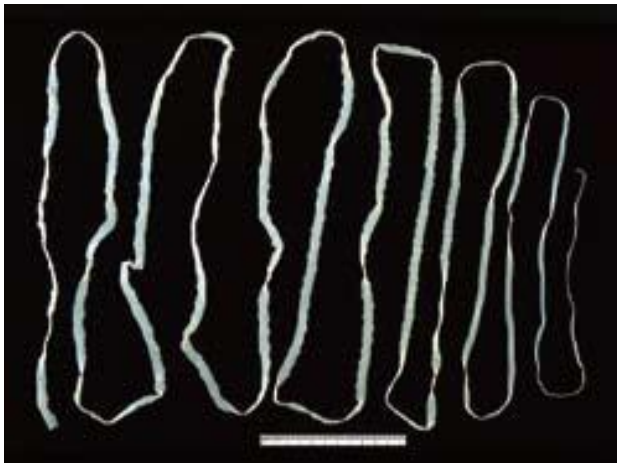
##A##

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Skill assessed	Mark
Analysing data	1

|!|BCEMC0200200059|!|

The following photograph shows a tapeworm that lives in the human intestine.



Which of the following statements about the effect of tapeworm on humans is correct?

A. It often causes fatal diseases in humans.
B. It often digests food for humans.
C. It often absorbs nutrients from humans.
D. It often damages the lining of the intestine in humans.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0200200060|!|

Which of the following is the common feature of the hosts of all parasites?

A. They are all animals.
B. They supply nutrients to the parasites.
C. Parasites live inside their bodies.
D. They are harmed by the parasites.

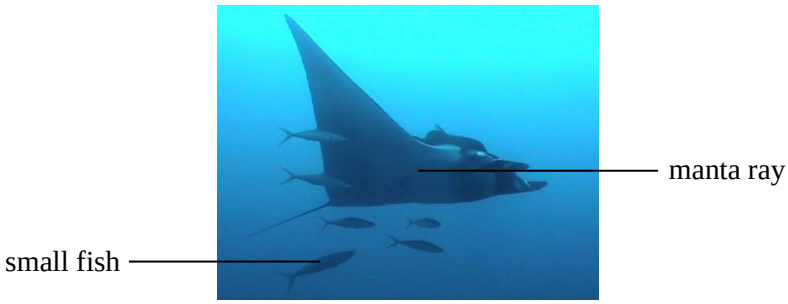
##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0200200061|!| ★

The photograph below shows a manta ray with some small fishes under its body. The fishes obtain food by removing the parasites from the manta ray's skin.



What is the ecological relationship between the manta ray and the small fish?

A. predation
B. parasitism
C. mutualism
D. commensalism

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0200200062|!| ★

Which of the following is ***not*** an example of mutualism?

A. Nitrogen-fixing bacteria live in the root nodules of leguminous plants.
B. An oyster is attached to the aerial roots of a mangrove plant.
C. Algae and fungi form lichens on a tree trunk.
D. Bacteria live inside the stomach or caecum of a ruminant.

##B##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0200500063|!|

✂

Directions: The two following questions refer to the different forms of nitrogen as shown below.

(1) nitrite
(2) ammonia
(3) nitrogen gas
(4) nitrate

Which form of nitrogen is the ***most*** abundant in the nitrogen cycle?

A. (1)
B. (2)
C. (3)
D. (4)

□

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0200500064|!|

✂

Which of the following conversions involves denitrifying bacteria?

A. (1) → (4)
B. (2) → (1)
C. (3) → (4)
D. (4) → (3)

□

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0200200065|!| ★

Pathogens (such as the bacteria that causes the plague) can be regarded as

A. hosts.
B. commensals.
C. parasites.
D. competitors.

□

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##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0200700066|!|

Which of the following can reduce impacts of human activities on ecosystems?

(1) use solar energy and reduce reliance on fossil fuels
(2) release pet animals into the wild
(3) restrict fishing activities during fishing moratorium

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0200700067|!|

Which of the following measures can prevent the extinction of rhinoceros?

(1) set up conservation areas for rhinoceros
(2) prohibit the hunting of rhinoceros
(3) increase public awareness by promoting trade of rhinoceros products

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0200200068|!| ★

The photograph below shows a colony of penguins.



Which of the following is an / are the adaptation(s) of the penguins to their habitat?

(1) They have a thick layer of feathers that covers their skin.
(2) They hibernate during the cold winter.
(3) They have a thick layer of subcutaneous fat.

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0200200069|!|

The following shows the different stages of primary succession.

(1) A climax community of woodland is established.
(2) Soil becomes suitable for herbaceous plants to grow.
(3) Spores of mosses are carried to the bare rocks by wind.
(4) Soil can support the growth of shrubs.
(5) Mosses become the dominant species.

Which of the following shows the correct sequence of the stages of primary succession?

A. (3) → (2) → (5) → (4) → (1)
B. (3) → (5) → (2) → (4) → (1)
C. (5) → (3) → (2) → (4) → (1)
D. (5) → (3) → (4) → (2) → (1)

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##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0200200070|!| ★

Which of the following combinations about the initial stages of primary succession and secondary succession is ***incorrect***?

	Primary succession	***Secondary succession***
A.	river deltas	land cleared by fire
B.	hardened volcanic lava flow	forest destroyed by logging
C.	rock exposed by landslide	shore disrupted by tsunami
D.	farmland destroyed by typhoon	new island formed after earthquake

□

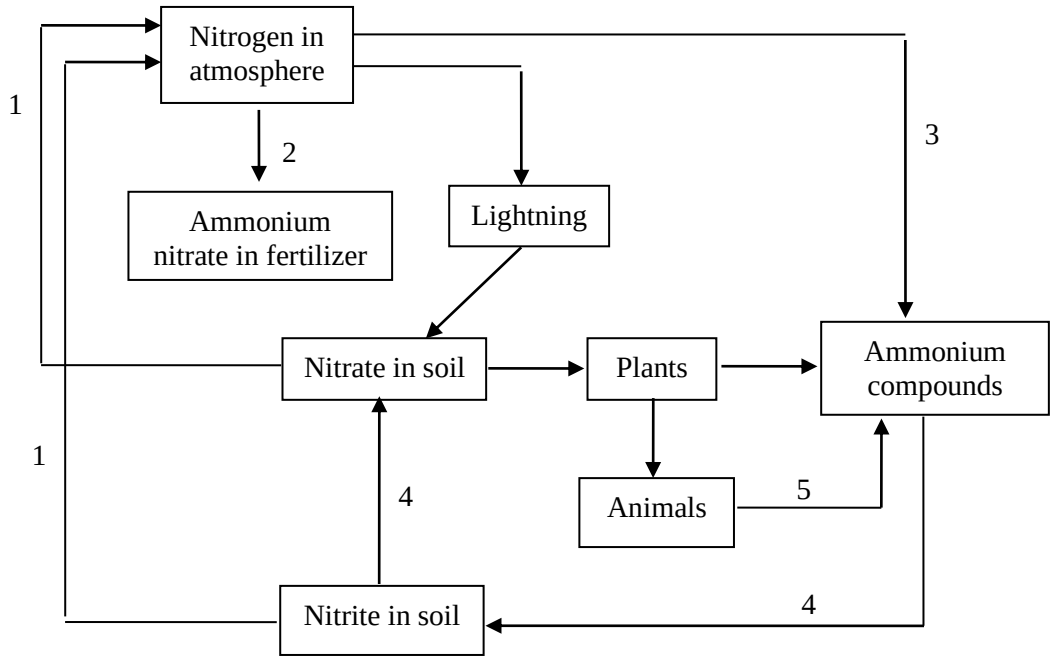
##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0200500071|!|

✂

The diagram below shows a nitrogen cycle.



Which of the following combinations correctly identifies the processes represented by the arrows?

	Denitrification	***Nitrogen fixation***	***Nitrification***
A.	1	3	4
B.	1	5	4
C.	2	4	5
D.	3	1	2

□

##A##

Skill assessed	Mark
Analysing data	1

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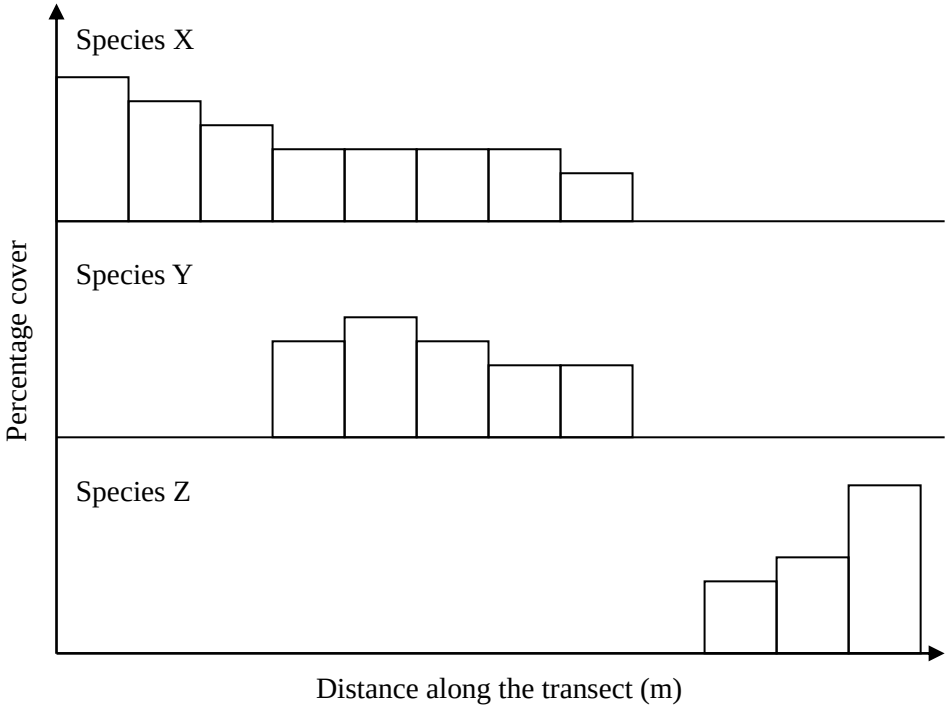
|!|BCEMB0200800072|!| ★

Peter and his classmates used a belt transect to investigate various plant distributions along a slope. **Graph I** shows the longitudinal section of the slope. **Graph II** shows the percentage cover of each species.

Graph I



Graph II



Which of the following ***cannot*** be concluded from the graphs above?

A. Species X is the most abundant at higher altitudes.
B. Species Z can only be found at heights below 35 m.
C. Species Y is the dominant species along the slope.
D. Species Y is evenly distributed across the relatively flat areas of the slope.

□

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##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0200800073|!|

Which of the following parameters can be measured with a refractometer?

A. wind speed
B. light intensity
C. salinity
D. relative humidity

□

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0200200074|!|

Which of the following comparisons between primary succession and secondary succession is ***incorrect***?

	Primary succession	***Secondary succession***
A.	starts without the prior presence of pioneer species	starts with pioneer species already established
B.	is more common	is less common
C.	usually starts without existing soil	starts with soil already present
D.	occurs more slowly	occurs more rapidly

□

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0200800075|!|

Which of the following comparisons between quadrat and line transect is correct?

	Quadrat	***Line transect***
(1)	suitable for habitats with fairly uniform conditions	suitable for habitats with significant variations in species distribution
(2)	measures the range of species distributions	measures the abundance of species
(3)	random sampling	systematic sampling

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0200100076|!|

Which of the following types of ecosystems in Hong Kong supports the greatest number of species?

A. grassland
B. woodland
C. mangrove
D. rocky shore

##B##

Skill assessed	Mark
Understanding basic concepts	1

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